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A review on multi-omics integration for aiding study design of large scale TCGA cancer datasets

Eonyong Han^{1†}, Hwijun Kwon^{1†} and Inuk Jung^{1*}

Abstract

Background Rapid advancements in high-throughput sequencing technologies allow for detailed and accurate measurement of omics features within their biological context. The integration of different omics types creates heterogeneous datasets, presenting challenges in analysis due to variations in measurement units, sample numbers, and features. Currently, there is a lack of generalized guidelines for making decisions in multi-omics study design (MOSD), such as selecting an appropriate number of samples and features, type of preprocessing and integration for robust analysis results. We propose a suggestive guideline for MOSD, involving nine important factors: sample size, feature selection, preprocessing strategy, noise characterization, class balance, number of classes, cancer subtype combination, omics combination, and clinical features.

Results To assess the effectiveness of our proposed MOSD guidelines, we designed and conducted seven benchmark tests using 10 clustering methods on various TCGA cancer datasets with an objective of clustering cancer subtypes. The results indicated robust performance in terms of cancer subtype discrimination when adhering to the following criteria: 26 or more samples per class, selecting less than 10% of omics features, maintaining a sample balance under a 3:1 ratio, and keeping the noise level below 30%. Feature selection was particularly important, improving clustering performance by 34%.

Conclusion These findings provide evidence-based recommendations for MOSD, enabling researchers to optimize analytical approaches and enhance the reliability of results across cancer datasets. The proposed MOSD framework offers a suggestive guideline addressing both computational and biological factors for multi-omics data integration.

Keywords Multi-omics, Integration, Study design, Machine learning

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Introduction

Recent advances in high-throughput sequencing and omics technologies have enabled the generation of extensive multi-omics data, proving valuable in identifying causal relationships for deeper understanding of biological functions [1–3]. Multi-omics integration (MOI) involves the simultaneous analysis of genomics, transcriptomics, epigenomics, proteomics, and metabolomics and has substantially advanced investigations into the molecular underpinnings of complex diseases such as cancer [4–8].

Various MOI tools have been developed for specific combinations of omics data, including IGC [9], PLRS [10], Oncodrive-CIS [11], MethyLMix [12], MMIA [13], MethGET [14], and linkedOmics [15]. Tools, such as, MOGSA [16], ActivePathways [17], multiGSEA [18], and iPanda [19] have also emerged to facilitate the interpretation of complex multi-omics results. Large-scale multi-omics archives, such as, The Cancer Genome Atlas (TCGA) [20], International Cancer Genome Consortium (ICGC) [21], Cancer Cell Line Encyclopedia (CCLE) [22], and Clinical Proteomic Tumor Analysis Consortium (CPTAC) [23] provide extensive multi-omics data with annotated clinical labels across different cancer types. However, leveraging this integration effectively poses numerous analytical and computational challenges. From a computational perspective, harmonizing distinct data types requires specialized approaches, as different omics data types can have varying distributions and sources of noise [24]. For instance, transcript expression follows a binomial distribution, while CpG islands associated with methylation display a bimodal distribution [25]. Biologically, different omics layers may produce complementary but occasionally conflicting signals, as demonstrated in studies of colorectal carcinomas where methylation profiles were linked to genetic lineages defined by copy number alterations (CNAs), but transcriptional programs showed inconsistent connections to subclonal genetic identities [26].

Previous research has identified several critical challenges in multi-omics integration, including the curse of dimensionality, data heterogeneity, missing data, class imbalances, and scalability [27]. Various studies have investigated specific aspects of these challenges - from sample size and noise variation effects on clustering performance [28] to the importance of feature selection [29]. Although these studies offer valuable insights into components of multi-omics analysis, none provide a unified framework that systematically addresses both computational and biological complexities across a broad range of factors [24, 28–31]. The extension of MOI analysis to new multi-omics datasets with smaller size or different omics types presents additional challenges. Questions persist about optimal feature selection, such as determining the

number of genes or omics features to use, and establishing minimum sample size requirements for qualitative analysis.

To address these gaps, we propose a structured suggestive guideline for Multi-omics Study Design (MOSD). This guideline synthesizes and extends previous work to identify nine key factors across computational and biological domains. Through comprehensive benchmarking across multiple TCGA datasets using ten clustering-based analytical methods, we evaluate how these factors affect MOI analysis across 10 cancer types. Our goal is to provide clear, evidence-based recommendations for researchers engaged in multi-omics research, offering minimal guidelines for designing multi-omics experiments and performing downstream analysis.

Materials and methods

Definition of MOSD factors

Through comprehensive literature review and systematic analysis, nine critical factors were identified that fundamentally influence multi-omics integration outcomes. This framework originated from Mirza et al. [27], who initially identified five key challenges in MOI analysis: the curse of dimensionality, data heterogeneity, missing data, class imbalances, and scalability. Building upon these foundations and incorporating insights from subsequent benchmark studies, these factors were organized into computational and biological aspects to provide a more complete analytical framework.

The computational factors begin with sample size, which addresses both the curse of dimensionality and scalability challenges. Chauvel et al. [28] demonstrated through both simulation and TCGA breast cancer data how clustering performance changes according to sample numbers, evaluating two clustering methods using the adjusted rand index (ARI) across mRNA, miRNA, methylation and protein omics data from 348 samples. This finding was further supported by Tini et al. [24], who evaluated five unsupervised algorithms using the F-measure method across multiple datasets including simulation data, murine liver data (BXD), and breast cancer gene expression data. Feature selection is a process where genes or omics features that are likely related to some biological trait of interest are carefully chosen by some algorithm, which is a standard procedure in most omics analysis. Pierre-Jean et al. [29] evaluated 13 unsupervised MOI methods for measuring the impact of feature selection. The study used three distinct simulated datasets (Gaussian, binary, and beta-like omics representations) to examine how the proportion of relevant variables impacts clustering performance. Tini et al. [24] also investigated feature selection's impact, though their filtering method based on coefficient of variation showed limitations in determining the optimal number of meaningful features

Table 1 Related study analysis

Study	Computational factors				Biological factors			Datasets	Omics	Method	Estima- tion method
	Sam- ple size	Feature selection	Preprocessing	Noise	Balance	Number of classes	Cancer subtype combination				
Chauvel et al. [28]	S	S	X	S	X	S	X	BRCA (348)	GE	BCC	ARI
									ME	MDI	
									MI	iCluster	
									Protein	moCluster	
										JIVE	
Rappoport et al. [30]	X	X	X	X	X	O	X	AML (170)	GE	iNMF	Survival analysis (logrank)
										k-means	Clinical distribution
								BIC (621)	ME	Spectral	
								COAD (220)	MI	LRacluster	
Tini et al. [24]	S	O	X	O	X	S	O	GBM (274)		PINS	
								KIRC (183)		SNF	
								LIHC (367)		rMKL-LPP	
								LUSC (341)		MCCA	
								SKCM (448)		MultiNMF	
								OV (287)		iClusterBayes	
								SARC (257)			
								BXD (66)	GE	MCCA	F-measure
								Platelet (12)	ME	JIVE	
								BRCA (491)	MI	MCIA	
									MFA		
									SNF		

Table 1 (continued)

Study	Computational factors				Biological factors			Datasets	Omics	Method	Estimation method		
	Sam-ple size	Feature selection	Preprocessing	Noise	Balance	Number of classes	Cancer subtype combination					Omics combination	Clinical feature
Pierre-Jean et al. [29]	S	S	S	O	S	S	X	O	O	BXD (64) Obesity (13) Liver cancer (360)	GE ME CNV Mutation	RGCCA intNMF SNF LRACluster PINSPlus CC MCIA mixkernel SGCCA iClusterPlus MoCluster CIMLR MOFA	F-measure ARI Survival analysis
	X	O	X	O	X	O	X	O	X	ACC (77) BRCA (760) COAD (291) KIRP (273) KIRC (314) LIHC (364) LUAD (453) LUSC (363) THYM (119)	GE ME MI CNV	SNF NEMO CIMLR MultiNMF PFA LRACluster moCluster iClusterBayes PINS Subtype-GAN	Precision NMI ARI F-measure

for class distinction. Preprocessing and noise characterization of multi-omics data represent vital computational factors. Duan et al. [31] demonstrated this in their benchmark of 10 MOI methods across nine TCGA cancer datasets, where they evaluated robustness by adding Gaussian noise at five different variance levels. Their work emphasized the importance of measuring both noise intensity and its distribution across the dataset. Sample balance across classes, initially identified by Mirza [27], was thoroughly examined by Rappoport et al. [30] in their benchmark of nine MOI methods across 10 TCGA datasets, where they evaluated clustering performance using both survival differences and clinical label correlations.

The biological factors incorporate cancer subtype combinations and multi-omics layer integration. Rappoport et al. [30] evaluated this through two measures: scoring survival differences of divided clusters as log ranks and analyzing 41 clinical labels using chi-square and Kruskal–Wallis tests. Duan et al. [31] further explored this aspect by testing 11 different combinations from four omics types to identify optimal configurations for individual and overall cancer subtyping, while also evaluating clustering accuracy, clinical significance, and computational efficiency. The four omics types were gene expression (GE), miRNA (MI), methylation (ME) and copy number variation (CNV). The framework also emphasizes clinical feature correlation, incorporating molecular subtypes, gender, pathological stage, and age. This aspect was particularly highlighted in Pierre-Jean et al.'s work [29] with real datasets (BXD, liver cancer, obesity), though their study noted limitations in applying simulation-derived factors to real clinical data. The framework addresses several limitations identified in previous studies. While Chauvel et al. [28] only performed single tests on TCGA-BRCA data, and Rappoport et al. [30] couldn't fully grasp the conditions needed to improve MOI method performance, this approach provides a more comprehensive evaluation. Similarly, it addresses the limitations noted in Tini et al.'s work [24] regarding noise application and feature selection, and expands upon Duan et al.'s analysis [31] by incorporating additional data conditions such as size and class balance.

Based on the comprehensive review of previous studies (Table 1), nine factors were identified and categorized into computational and biological factors. The computational factors include: (1) sample size, (2) feature selection, (3) preprocessing strategy, (4) noise characterization, (5) class balance and (6) number of classes. The biological factors comprise: (7) cancer subtype combinations, (8) omics combinations, and (9) clinical feature correlation. These factors provide a structured framework for evaluating and improving multi-omics data integration methods. By consolidating the efforts of previous research and addressing their limitations, we aimed

to provide a more complete analytical approach that enhances both the precision and reproducibility of multi-omics analyses. The validation process developed specific benchmarks for each factor and applied them across various TCGA cancer datasets, allowing for comprehensive assessment of their impact on multi-omics integration outcomes.

Data acquisition and assembly

A comprehensive analysis of data sourced from TCGA [20] was used to provide the suggestive MOSD guidelines. This study utilized multi-omics data from TCGA repository, focusing on 3,988 patients across ten cancer types: bladder urothelial carcinoma (BLCA, $n=400$), breast invasive carcinoma (BRCA, $n=592$), colon adenocarcinoma (COAD, $n=249$), head and neck squamous cell carcinoma (HNSC, $n=471$), kidney renal papillary cell carcinoma (KIRP, $n=268$), liver hepatocellular carcinoma (LIHC, $n=357$), lung adenocarcinoma (LUAD, $n=442$), skin cutaneous melanoma (SKCM, $n=350$), stomach adenocarcinoma (STAD, $n=365$), and thyroid carcinoma (THCA, $n=494$). These cancer types were selected based on the availability of patient-matched omics and clinical data, with a minimum threshold of 100 patients per cancer type to ensure robust statistical analysis. Detailed breakdown of subtype and sample counts is listed in Supplementary Table S1.

As shown in Table 2, the multi-omics layers incorporated in this study comprised gene expression (GE), miRNA (MI), mutation data, copy number variation (CNV), and methylation (ME), representing the commonly utilized data types in multi-omics research [32]. The genomic features varied across cancer types, with GE features ranging from 2,097 to 2,229, while LIHC demonstrated the most extensive profile with 17,390 GE features and 21,933 CNV features. Notably, mutation data was particularly comprehensive for LIHC, encompassing 396,065 features. Figure 1 shows the structure of the multi-omics data, which were divided to omics and clinical feature data types for the analysis.

Clinical annotations were systematically integrated, including molecular subtypes, gender distribution, pathological staging, and age demographics. The molecular subtypes for each cancer type were extracted from TCGA clinical data and varied by cancer type, with some cancers showing distinct classification system. For instance, for BRCA, we applied PAM50-based subtypes [33] (Luminal A, Luminal B, Basal, HER2), and for COAD, Consensus Molecular Subtype 1–4 (CMS1–4) were classified based on Guinney et al. [34]. For other cancer types, including LUAD, STAD, SKCM, KIRP, and HNSC, we utilized the 'Subtype' column provided in the TCGA clinical matrices, which included both categorical descriptors and numeric classifications (ranging 1–6 depending on

Table 2 Statistics of the TCGA dataset

Cancer	Number of omics features				Number of samples for each clinical feature			
	GE	ME	MI	CNV	Subtype	Pathologic-stage	Gender	Age
BLCA	17390	396065	2195	21933	1 (41) 2 (41) 3 (29) 4 (15) Total (126)	Stagell (128) StageIII (136) StageIV (132) Total (396)	MALE (296) FEMALE (104) Total (400)	Middle (124) Old (244) Total (368)
BRCA			2222		LumA (323) LumB (118) Basal (102) Her2 (39) Total (582)	Stagel (96) Stagell (345) StageIII (151) Total (592)	-	Young (16) Middle (221) Old (81) Total (318)
COAD			2097		CMS1 (34) CMS2 (67) CMS3 (35) CMS4 (63) Total (199)	Stagel (36) Stagell (83) StageIII (67) StageIV (29) Total (215)	MALE (137) FEMALE (112) Total (249)	Middle (99) Old (130) Total (229)
HNSC			2229		Basal (74) Mesenchymal (65) Atypical (59) Classical (44) Total (242)	Stagel (26) Stagell (71) StageIII (74) StageIV (234) Total (405)	MALE (347) FEMALE (124) Total (471)	Middle (261) Old (168) Total (429)
KIRP			2099		C1 (84) C2a (34) C2b (21) Total (139)	Stagel (164) Stagell (18) StageIII (49) StageIV (14) Total (245)	MALE (200) FEMALE (68) Total (268)	Middle (141) Old (91) Total (232)
LIHC			2157		1 (65) 2 (55) 3 (63) Total (183)	Stagel (166) Stagell (84) StageIII (61) Total (311)	MALE (243) FEMALE (114) Total (357)	Middle (171) Old (129) Total (300)
LUAD			2211		1 (17) 2 (22) 3 (38) 4 (26) 5 (43) 6 (33) Total (179)	Stagel (233) Stagell (108) StageIII (62) StageIV (19) Total (422)	MALE (205) FEMALE (237) Total (442)	Middle (174) Old (212) Total (386)
SKCM			2203		BRAF (116) RAS (78) TWT (37) NF1 (23) Total (254)	Stagel (72) Stagell (67) StageIII (137) StageIV (20) Total (296)	MALE (220) FEMALE (130) Total (350)	Middle (175) Old (103) Young (29) Total (307)
STAD			2161		CIN (202) MSI (59) GS (46) EBV (29) Total (336)	Stagel (46) Stagell (116) StageIII (163) StageIV (28) Total (353)	MALE (240) FEMALE (125) Total (365)	Old (177) Middle (133) Total (310)
THCA			2201		1 (131) 2 (67) 3 (82) 4 (105) 5 (90) Total (475)	Stagel (279) Stagell (50) StageIII (109) StageIV (46) Total (484)	MALE (133) FEMALE (361) Total (494)	Middle (237) Young (166) Old (45) Total (448)

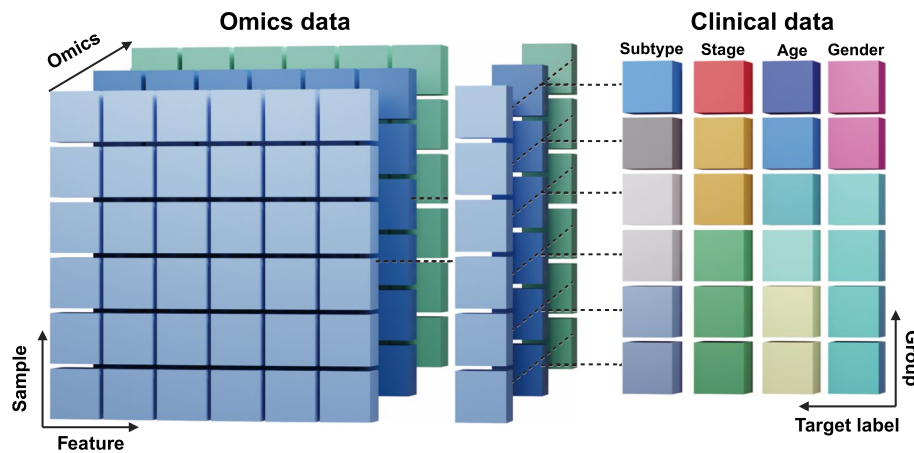


Fig. 1 Structure and organization of multi-omics data integration. Multi-omics datasets are structured as a collection of two-dimensional matrices, where each matrix represents a distinct omics layer (e.g., genomics, transcriptomics, proteomics). Rows consistently represent biological samples across all matrices, while columns contain features specific to each omics type (e.g., genes, proteins, methylation probes). A complementary clinical matrix maintains the same sample-wise organization (rows) but contains relevant clinical variables (columns) such as age, sex, and disease status. This standardized structure enables integrated analysis across multiple biological layers while maintaining sample correspondence

cancer type). These subtype labels were used as-is for benchmarking purposes without reclassification or biological reinterpretation. Age stratification into Young, Middle, and Old categories demonstrated a predominance of middle-aged and elderly patients, with specific age-related patterns varying by cancer type. The University of California Santa Cruz (UCSC) Xena browser [35] was used for collecting the multi-omics data. Patient barcodes were verified through TCGAbiolinks [36] to ensure data matching.

Data preprocessing

The examination of preprocessing effects on integration performance involved the creation of three distinct dataset types, each representing different levels of data refinement. The first category comprises Raw Datasets (RAW), which maintain data in its most basic processed form through log2 transformation of the original values. This approach preserves the inherent structure of the data while enabling basic computational analysis. The second category consists of Normalized Datasets (NORM), which underwent quantile normalization followed by 0-1 scaling. This process establishes uniform distributions across various omics layers, facilitating more standardized comparisons.

The third category encompasses Gene Level Datasets (GL), representing an advanced preprocessing approach where normalized data aligns with gene expression characteristics. To enable integrative analysis across heterogeneous omics layers, we converted all omics data into a gene-level format, following a gene-centric integration strategy as described in Jung et al. [37] For methylation, we selected probes located within a 2 kilobase upstream region from the promoter region of each gene. The

average beta value across these probes was computed per gene. miRNA expression values were assigned to their corresponding target genes based on experimentally validated miRNA–gene interactions, and the geometric mean of all miRNAs targeting a gene was used as the representative value. For CNV, segmented values overlapping gene bodies were averaged per gene. After alignment across omics layers, each matrix was min–max scaled to a 0–1 range to ensure comparability and avoid dominance from any single data type during integration. This process ensures that all omics matrices share the same gene and sample dimensions, enabling structured gene-level multi-omics representation. The complete preprocessing methodology and associated code reside in the project's GitHub repository, ensuring transparency and reproducibility.

The preprocessing strategy specifically addressed the challenge of missing data, which previous research has identified as detrimental to multi-omics analyses. To ensure consistency across omics layers and avoid the influence of missing values, we adopted a conservative approach by strictly excluding any omics features that contained missing entries. In addition, sample groups with fewer than ten individuals were removed to maintain sufficient representation for statistical analysis. No imputation was applied during preprocessing. Such measures maintain robust representation across categories while minimizing potential biases stemming from data sparsity.

The processing of clinical data involved careful categorization to enable meaningful correlation with multi-omics findings. The framework retained categorical features including molecular subtype, gender, and pathological stage as natural labels for subsequent analyses.

Age data underwent transformation from its continuous form into three discrete categories: young (0-39 years), middle aged (40-64 years), and older (65 years and above), following protocols in cancer research literature [38]. This age categorization facilitates the investigation of age specific molecular patterns within each cancer type [39].

MOI analysis methods and implementation

We evaluated ten MOI methods that employ various computational approaches including clustering, matrix factorization, and network integration across ten TCGA cancer datasets to determine optimal conditions for cancer data analysis. These methods were selected based on three criteria: ease of implementation in R or Python, prevalence in academic research, and effectiveness with default parameters. Rather than focusing solely on comparison of method performance, our analysis examined how each method's characteristics interacted with the nine factors through seven benchmark tests. Through a series of controlled benchmark tests, we identified parameter conditions that consistently improved analysis quality under each method, which we refer to as analytically favorable or recommended setups for multi-omics study.

Similarity network fusion

Similarity Network Fusion (SNF) [40] constructs similarity networks for each omics type using metrics such as Pearson correlation or chi-square and then fuses these networks into a unified representation. Its advantage in MOSD lies in integrating disparate omics layers, handling clinical features of both discrete and continuous types, and tolerating small sample sizes.

Spectrum

Spectrum [41] is a density-aware spectral clustering method that leverages eigenvectors from the multi-omics sample-graph matrix to mitigate noise and illuminate biologically relevant structures. It uses a k-nearest neighbors kernel, obviating the need for extensive parameter tuning across varied datasets. Spectrum's advantages for MOSD include scalability to large datasets and the ability to capture subtle density variations, aided by a Gaussian Mixture Model (GMM) rather than traditional k-means.

PINSPlus

PINSPlus (PP) [42], an extension of PINS [43], integrates multi-omics data by building connectivity graphs for each omics layer and fusing them to identify stable clusters. PP showed to be able to find both known and unknown subtypes with significant clinical implications.

Neighborhood-based multi-omics clustering

Neighborhood-based Multi-omics Clustering (NEMO) [44] is a flexible algorithm that operates through three main steps: 1) calculation of pairwise distances between samples within each omics layer using specified distance metrics, 2) construction of k-nearest neighbor graphs for each sample based on local neighborhood patterns, and 3) clustering of samples using graph-based clustering algorithms applied to these neighborhood representations. NEMO supports various feature types including numerical, discrete, ordinal, and imaging data, and excels with partial datasets by eliminating the need for missing data imputation while maintaining faster execution compared to other multi-omics clustering methods.

Cluster of cluster assignments

Cluster of Cluster Assignments (COCA) [45] converts each omics dataset into a one-hot encoded cluster matrix and then applies consensus clustering, typically using Pearson's correlation to form a similarity matrix. The consensus clustering provides stability and helps identify cross-omics correlations without strict normalization requirements.

Low-rank approximation clustering

Low-Rank Approximation Clustering (LRAcluster) [46] adopts a probabilistic model that treats omics data as Gaussian-distributed variables under a low-rank constraint. It compresses high-dimensional multi-omics data into a more interpretable, low-dimensional representation. LRAcluster accurately detected shared covariance structures across omics types and highlighted critical regulatory signatures and cancer subtypes.

Consensus clustering

Consensus Clustering (CC) [47] repeatedly applies clustering algorithms to resampled versions of the dataset and aggregates these results into a consensus solution, mitigating random fluctuations. This is particularly valuable in MOSD when handling high-dimensional and heterogeneous data. By quantifying cluster assignment frequencies, CC not only consolidates robust groupings but also reveals the stability of potential cluster solutions across multiple resampling iterations.

Multi-omics factor analysis

Multi-Omics Factor Analysis (MOFA) [48] decomposes multi-omics data into latent factors that capture the main source of variation across different layers. By fitting a linear factor model, it effectively integrates incomplete data, highlights outliers, and identifies molecular signatures driving heterogeneity. MOFA's interpretability and capacity for imputing missing values offer considerable benefits

within MOSD, allowing deeper insight into latent structures that span multiple data types.

iClusterPlus

iClusterPlus [49] is an advanced version of iCluster [50] that integrates multi-omics data using a probabilistic joint latent-variable model. The method incorporates different omics data types (binary, categorical, and continuous) into a low-dimensional latent representation through estimation of latent variables W and H that capture relationships between samples and features. Model optimization employs a lasso penalty (L1-norm) on H to obtain sparse solutions, enabling dimension reduction by removing uninformative features. This approach improves clustering performance while increasing interpretability and computational efficiency through comprehensive multi-omics integration.

Integrative non-negative matrix factorization

Integrative Non-negative Matrix Factorization (IntNMF) [51] extends classic NMF [52] by sharing a common basis matrix W across multiple two-dimensional omics datasets, while allowing each data type to have a distinct coefficient matrix H . It enhances the detection of subtypes and leverages resampling-based cross-validation for selecting the optimal number of clusters. IntNMF's strength lies in its ability to handle large-scale datasets without relying on strict distributional assumptions. Collectively, the ten methods offer a wide range of strategies for multi-omics integration and clustering, each with unique advantages for different data structures and MOSD factors. By systematically benchmarking their performance across TCGA cancer datasets, we aimed to pinpoint favorable conditions, such as sample size requirements, handling of missing values, or optimal omics combinations, that maximize analytical precision and biological relevance under the MOSD framework.

Benchmark design

To address the heterogeneous nature of multi-omics data and evaluate parameter influence on analysis outcomes, a series of comprehensive benchmarks aligned with the previously defined computational and biological factors in MOSD has been developed. Each benchmark test isolated a single study design factor while maintaining all other variables at fixed default settings. This allowed fair comparison of individual factor impacts on clustering performance. The full test design, including parameter ranges, evaluation objectives, target values, and default settings for each benchmark, is summarized in Table S2. This systematic isolation of variables allowed for precise measurement of each factor's unique contribution to analysis outcomes. The evaluation process employed multiple performance metrics, including Normalized

Mutual Information (NMI), Adjusted Rand Index (ARI), and the F-measure. To ensure statistical robustness, each benchmark underwent ten iterations with random sample selection.

Determining the minimal required sample size

A key objective in MOSD is to identify the minimal sample size required for reliable multi-omics clustering. In this test, we randomly selected subsets of increasing size beginning with as few as five samples to track when performance metrics plateaued. As illustrated in Fig. 2, this plateau reflects the threshold beyond which additional samples provide diminishing returns for accurate cancer subtyping, thus indicating the minimum cohort size needed for robust analysis.

Optimizing feature selection

Determining and adequate balance between feature quantity and quality is important in MOSD. To assess the effect of feature quantity on MOI performance, we incrementally selected features based on their relevance to sample group labels (i.e., cancer subtypes). Chi-square tests were applied for GE, MI, and ME omics types, while ANOVA was applied to CNV data due to its inclusion of negative values. Features were ranked by p -value, and top-ranked subsets were selected at increasing proportions—1% steps for early iterations, then 10%—up to 100% of features. This strategy was chosen over variance-based or classifier-specific ranking to avoid bias from model dependencies and to provide a group-informed yet algorithm-agnostic selection method. The exact number of features per omics layer in each benchmark iteration is provided in the GitHub repository. The proportion of selected features in each scenario is shown in Fig. 2.

Measuring the impact of different preprocessing methods

Preprocessing can significantly affect the outcomes of multi-omics analysis. We compared three approaches: RAW, NORM, and GL. RAW data underwent only log2 transformation, while NORM underwent quantile normalization and min-max scaling. GL integration averaged omics features at the gene level, as recommended by Jung et al. [47, 53] potentially enhancing gene-centric signals yet discarding non-gene-aligned features. By applying these variants to the same sets of MOI methods, we examined whether specific preprocessing strategies yield consistently better clustering performance (Fig. 2).

Evaluating class balance

Class imbalance, where certain cancer subtypes or clinical categories contain fewer samples, can bias clustering and reduce the detection of smaller classes [54]. In this benchmark, we progressively introduced imbalance in

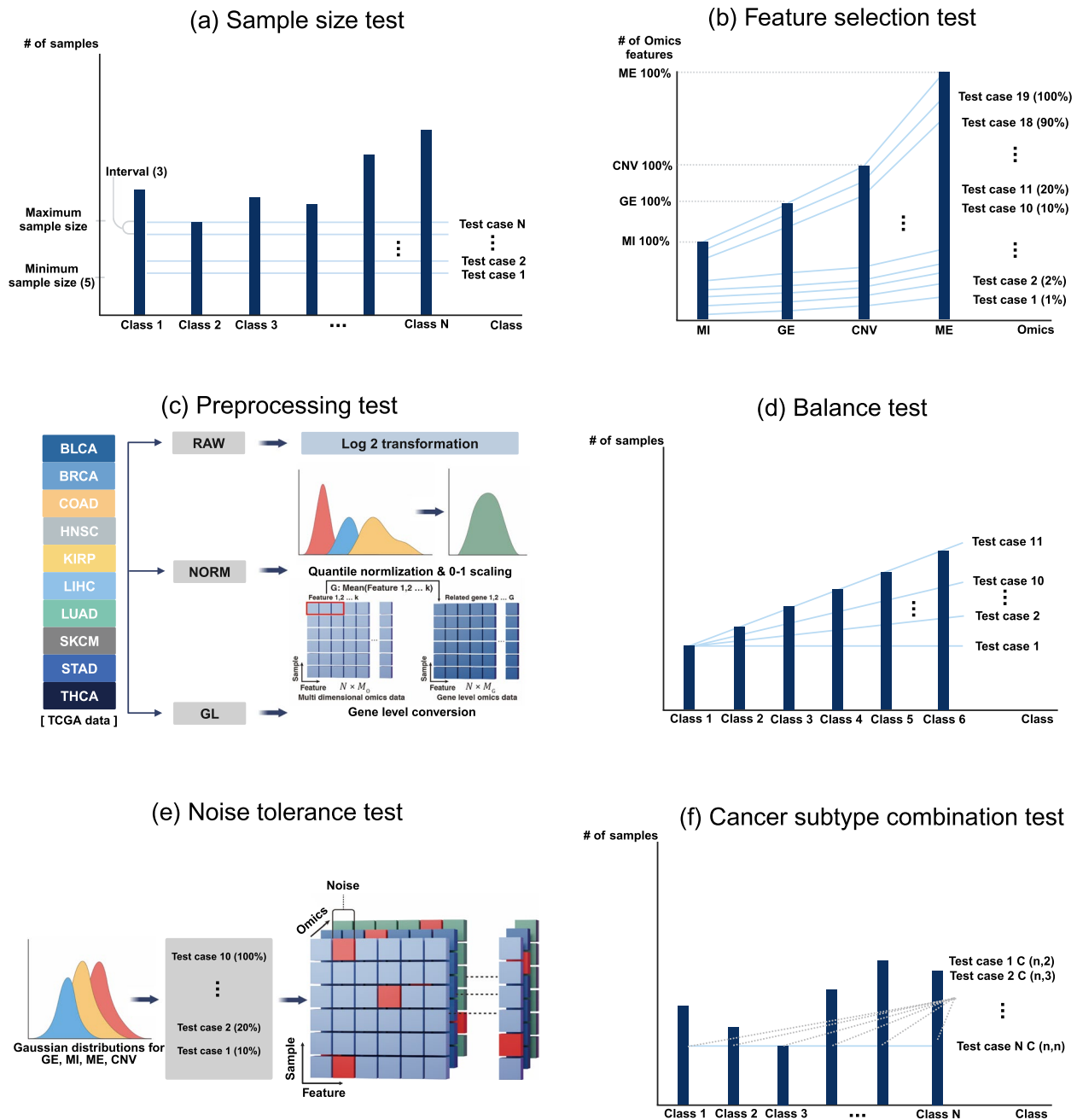


Fig. 2 Benchmarking designs of key factors. Systematic evaluation of six critical factors in multi-omics integration: **a** Sample size - Performance metrics with increasing sample sizes showing minimum cohort requirements. **b** Feature Selection - Impact of varying feature numbers selected via statistical testing across omics types. **c** Preprocessing test - comparison of RAW (Log2 transformed), NORM (Quantile normalization with 0-1 scaling), and GL (Gene level dataset) approaches on clustering performance. **d** Class balance - effects of classes with imbalanced sample on method accuracy. **e** Noise tolerance - Method resilience to incremental Gaussian noise (10-100%). **f** Cancer subtype combination - Performance analysis across different cancer subtype combinations. Each panel demonstrates the parameter's influence on MOI performance

sample sizes to gauge how well different MOI methods cope with skewed distributions (Fig. 2).

Assessing noise tolerance

Multi-omics datasets often include noise, or errors, from technical and biological reasons [29]. To evaluate its influence, Gaussian noise was introduced in 10%

increments up to 100% in each omics layer (Fig. 2). The objective was to observe the maximum tolerable noise threshold where clustering performance started declining significantly.

Cancer subtype combinations

Different subtypes within the same cancer type frequently display unique molecular signatures. We explored whether varying combinations of these subtypes influence clustering performance, focusing on how distinguishable they are (Fig. 2). All clustering evaluations were performed on datasets containing at least two distinct subtype labels, ensuring meaningful calculation of performance metrics. The benchmark also examined the number of subtypes assessed—ranging from a low count (2–3) to more diverse sets—to identify at which point distinguishing multiple subtypes becomes challenging or more susceptible to errors.

Omics combination

Finally, we investigated how specific omics-layer combinations affect clustering for four clinical features (molecular subtype, gender, pathological stage, age). Data configurations ranged from single-omics (GE alone) to multi-omics (e.g., GE-ME-CNV or GE-ME-MI-CNV). Although combining more omics can reveal intricate biological insights, it can also amplify heterogeneity, aligning with findings by Duan et al. [31] and Reel et al. [55]. Hence, determining which combinations yield the most accurate results for each clinical feature is a key focus in MOSD, and our test delineates situations where additional omics layers either enhance or degrade performance. Through these targeted benchmarks, we provide a systematic exploration of how each factor contributed to multi-omics analysis results. This approach offers actionable insights into the design, execution, and interpretation of multi-omics studies.

Performance evaluation for MOI methods

Evaluating the performance of clustering-based MOI methods is an essential component of the MOSD framework. Three metrics were used to evaluate the clustering performance, ARI, NMI, and F-measure [56]. ARI is calculated from the Rand Index (RI), which measures clustering accuracy:

$$RI = \frac{TP + TN}{TP + FP + TN + FN}$$

Here, TP represents the sample pairs correctly grouped within the same cluster, while TN represents the pairs correctly grouped in different clusters. FP indicates the pairs incorrectly grouped in the same cluster, and FN represents pairs incorrectly grouped in different clusters. ARI adjusts the RI by accounting for chance:

$$ARI = \frac{RI - Expected(RI)}{Max(RI) - Expected(RI)}$$

The Expected(RI) represents the expected RI value by random chance, while Max(RI) indicates the maximum possible RI value. NMI measures the mutual information between the true labels (T) and clustering results (C). The probability distributions are defined as:

$$P(i) = \frac{|T_i|}{N}$$

$$P'(j) = \frac{|C_j|}{N}$$

$$P(i, j) = \frac{|T \cap C|}{N}$$

Here, N represents the total number of elements, while $|T_i|$ indicates the number of elements in the true class i , and $|C_j|$ represents the number of elements in the predicted cluster j . The intersection $|T \cap C|$ denotes the number of elements shared between the true class and predicted cluster. The mutual information is then calculated as:

$$MI = \sum_{i \in T} \sum_{j \in C} P(i, j) \log \left(\frac{P(i, j)}{P(i) P'(j)} \right).$$

To normalize MI, we calculate the entropy of both true labels and clustering results. Entropy measures the uncertainty or randomness in the clustering assignments. The entropy for true labels and clustering results are calculated as:

$$H(T) = - \sum_{i \in T} P(i) \log(P(i))$$

$$H(C) = - \sum_{j \in C} P'(j) \log(P'(j))$$

where $P(i)$ is the probability of an element belonging to the true class i , and $P'(j)$ is the probability of an element being assigned to cluster j . Using these entropy values, we normalized MI to obtain NMI:

$$NMI = \frac{MI(T, C)}{mean(H(T), H(C))}$$

This ensures the NMI value falls between 0 and 1, where 1 indicates perfect clustering alignment with true labels, and 0 indicates completely random clustering. The average of the entropies in the denominator acts as a normalization factor, making NMI more comparable across different clustering scenarios. The F-measure computation involves three set operations:

$$A = |Pairs(U) \cap Pairs(V)|$$

$$B = |Pairs(U) - Pairs(V)|$$

$$C = |Pairs(V) - Pairs(U)|$$

The sets U and V represent the predicted clusters and true clusters respectively, where Pairs(X) generates all possible pairs of elements within set X. The F-measure is then calculated as:

$$F\text{-measure} = \frac{2A}{2A + B + C}$$

To obtain an overall performance metric, we compute a cluster-score as the arithmetic mean of all three metrics:

$$\text{cluster-score} = \frac{ARI + NMI + F\text{-measure}}{3}$$

Although ARI theoretically ranges from -1 to 1, all observed values in our benchmark were non-negative, eliminating the need for normalization before averaging with NMI and F-measure. This combined cluster-score serves as our primary evaluation metric, providing balanced assessment of clustering performance across all benchmark tests. Individual metrics are examined when detailed analysis is required. We note that future studies encountering negative ARI values should consider normalization to a 0–1 scale to ensure balanced integration into composite scoring.

Results

Optimal and suboptimal configurations in benchmark tests

Identifying the most and least effective parameter configurations were searched to characterize the importance of each of the nine key factors. Regarding sample size and class balance, smaller but balanced classes often performed better than larger but imbalanced ones, as exemplified in BLCA where 11 samples per class achieved the highest cluster-score (0.55), while very small classes (i.e., 5 samples) showed poor performance. Feature selection demonstrated that using fewer features generally yielded better results than using all features, which consistently produced the lowest scores across cancer types. The class balance benchmark showed varying optimal configurations across cancer types. Some cancers performed best with equal class sizes (e.g., BRCA with 40 samples per class), while others showed resilience to moderate imbalance. The noise tolerance test revealed that increasing noise levels universally degraded performance, with 100% noise consistently producing the lowest scores (often below 0.10) across all cancer types.

For biological factors, certain subtype combinations demonstrated notably stronger signals. For instance, Luminal B-Basal subtypes in BRCA and CIN-EBV in STAD showed particularly high discrimination (score >0.90). For omics combinations, GE-ME pairings frequently emerged as the most effective omics pairs across multiple cancer types, while combinations involving CNV often showed poorer performance. Notably, some tests, such as gender analysis in BRCA, were excluded due to extreme data imbalance, highlighting the importance of considering data distribution in study design.

These findings across different parameter configurations provide insights for optimizing multi-omics analysis approaches and aid in parameter selection. The complete detailed results are available in Supplementary Tables S3 and S4.

Impact of sample size

The relationship between sample size and clustering performance was examined across all cancer types (Fig. 3a). The analysis revealed that increasing sample size from 5 to 38 samples per class had a subtle but positive effect on average cluster scores, as indicated by the gradually rising trend line (red). However, the most notable impact of larger sample sizes was on score consistency rather than absolute performance - the violin plot distributions become narrower as sample size increases, particularly after 26 samples, indicating reduced variance in clustering performance. While the average cluster score shows only modest improvement (i.e., ranging approximately from 0.5 to 0.55), the minimum scores become notably higher with increased sample size, suggesting that larger sample sizes help prevent poor clustering outcomes. Detailed cancer-type specific trends are provided in Supplementary Figure S1.

Effect of feature selection

Feature selection appears to influence clustering accuracy, with certain tendencies observed in the relationship between feature proportion and performance (Fig. 3b). In the analyzed dataset, lower feature proportion (1–10% of features) tended to yield higher cluster scores, averaging around 0.55, albeit with a relatively wide distribution. As the feature proportion increased (20–50%), a gradual decline in performance was observed, followed by a further reduction at higher feature proportion (60–100%), where mean cluster scores approached 0.4. Notably, the distribution patterns change markedly across these phases. The lower feature proportion (1–10%) showed wider distributions, indicating greater variability in outcomes but also higher potential for optimal performance. The transition phase showed gradually narrowing distributions, suggesting more consistent but declining performance. The higher feature proportion (particularly

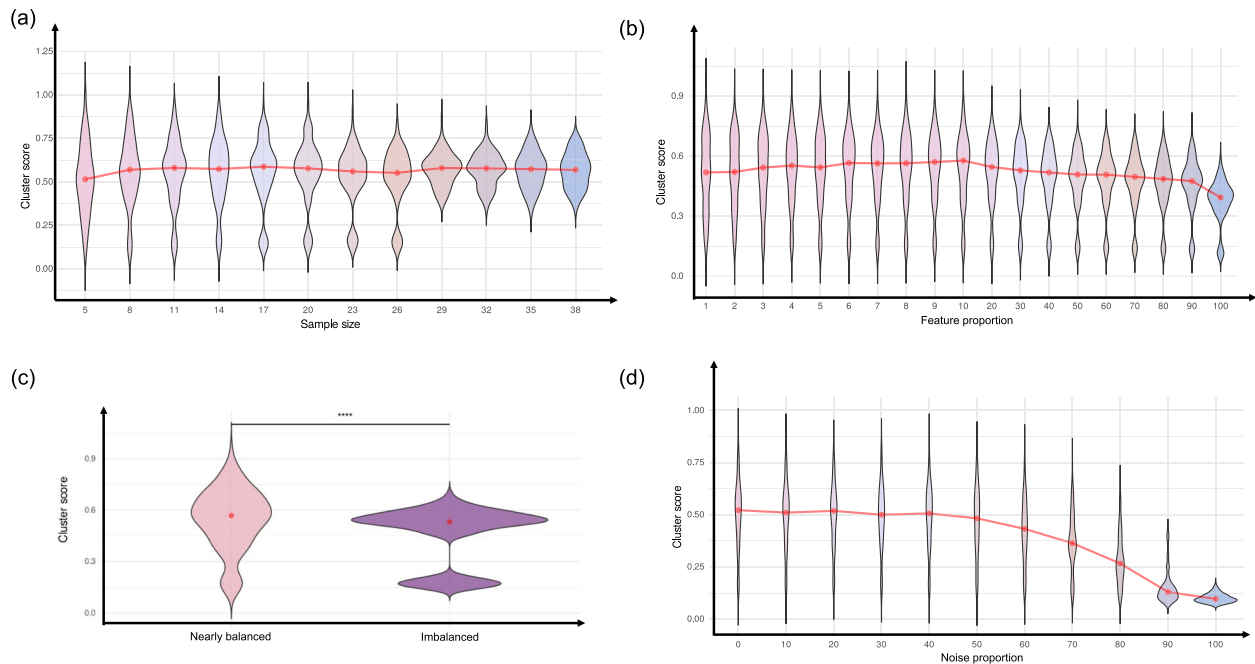


Fig. 3 Impact of 4 factors on Multi-omics clustering performance. Violin plots showing the distribution of clustering performance scores across different parameters: **a** Sample size effect - Performance trends with increasing sample sizes (5-38 samples per class), showing modest improvement in mean scores (red trend line) but notably decreased variance with larger samples. **b** Feature selection impact - The relationship between feature proportion and clustering performance, demonstrating optimal performance at 1-10% feature selection, decline at 20-50%, and consistent poor performance at 60-100%. **c** Class balance analysis - comparison of clustering performance between nearly balanced (<3:1 ratio) and imbalanced (>3:1 ratio) classes, showing significantly higher performance (**** $p < 0.0001$) in relatively balanced conditions. **d** Noise tolerance - distinct impact of performance response to increasing noise levels: stable (0-50%), decline (50-80%), and severe degradation (80-100%). Red lines indicate overall trends, and violin plot widths represent score distribution density at each parameter level

90-100%) showed both decreased performance and narrower distributions, indicating consistently poorer outcomes when using most or all features.

Such pattern suggests that selective feature usage, particularly under 20%, is recommended for optimal clustering performance. The sharp decline in performance with larger feature sets clearly indicates that selecting fewer but biologically relevant ones is beneficial. Detailed cancer-type specific trends are available in Supplementary Figure S1.

Preprocessing influence

The comparative analysis of three preprocessing approaches—RAW, NORM, and GL—revealed nuanced patterns across different cancer types and analytical contexts. While NORM datasets showed a slight overall advantage, the performance differences between methods were generally modest, typically below 0.1, suggesting robustness across the different preprocessing approaches. NORM showed strength in gender-based analyses, with consistently higher performance improvements, ranging from +0.20 to +0.27, across multiple cancer types including BLCA, COAD, HNSC, and LUAD. This suggests that normalization may be especially beneficial when analyzing gender-specific molecular patterns.

Cancer-type specific patterns were observed where some types showed consistent preferences for particular pre-processing methods. For instance, LUAD demonstrated better performance with GL preprocessing across multiple test categories, while STAD consistently favored NORM preprocessing, indicating that optimal preprocessing strategies may be cancer-type dependent.

Raw data preprocessing showed advantages in noise tolerance tests for several cancer types (notably BLCA, BRCA, and KIRP), potentially indicating that normalization may sometimes weaken the signals relevant to certain biological features. These findings suggest that while the choice of preprocessing can impact analysis outcomes, the effect is often context-dependent, varying by cancer type and analytical objective. The complete comparative analysis is available in Supplementary Table S5.

Influence of class balance

The analysis of sample balance showed distinct difference in clustering performance between nearly balanced and imbalanced classes (Fig. 3c). Datasets with nearly balanced classes (imbalance ratio < 3:1) yielded significantly higher and more consistent cluster scores compared to imbalanced sample classes (imbalance ratio > 3:1), as indicated by the statistically significant difference (adj. p

< 0.0001). Classes with nearly balanced samples showed not only higher cluster scores (i.e., 0.58) but also exhibited a wider range of potential high-performance outcomes, as illustrated by the broader upper portion of the violin plot. In contrast, classes with imbalanced samples demonstrated lower performance (0.52) and a distinctive bimodal distribution, suggesting two distinct performance patterns in imbalanced datasets. The narrower distribution in the imbalanced classes indicates more predictable, though generally lower, performance outcomes. Collectively, the overall reliability of clustering results is likely to decline when the sample ratio exceeds 3:1. Detailed cancer-type specific trends are available in Supplementary Figure S1.

Noise-induced effects on clustering outcomes

The analysis of noise tolerance in multi-omics integration showed distinct performance drop after a certain level of noise (Fig. 3d). The violin plots demonstrate a clear response to increased noise levels. The performance was stable with 0-50% noise where cluster scores maintained around 0.5 with consistent performance. The cluster score started to decline with 50% noise showing gradual decline. Interestingly, the variance in performance decreased substantially with increasing noise, suggesting that high noise levels not only degrade performance but also make outcomes more predictably poor. The sharp decline after 80% noise indicates a critical threshold beyond which reliable clustering becomes nearly impossible, regardless of the MOI method used. Detailed cancer-type specific responses to noise are available in Supplementary Figure S1.

Cancer subtype combination effect

Analysis of cancer subtype combinations across ten TCGA cancer types revealed distinct patterns in clustering performance (Fig. 4). Cancer-type-specific molecular characteristics significantly influence multi-omics clustering performance, with cluster-score varying across both cancer types and specific subtype combinations. In BRCA, Luminal A vs. Luminal B classification consistently showed lower performance due to their shared estrogen receptor positivity and overlapping expression profiles, differing primarily in proliferation-related genes [33, 57]. Similarly, COAD clustering of CMS2 and CMS3 subtypes yielded lower accuracy than more distinct pairs like CMS1 and CMS4, reflecting their shared epithelial characteristics and overlapping genomic features [34, 58]. These results demonstrate that clustering difficulty correlates with underlying molecular similarity between subtypes, emphasizing the importance of selecting biologically distinct subtypes for reliable integrative clustering outcomes. Method-specific performance patterns emerged across cancer types. SNF, CC, and Spectrum

methods consistently maintained higher performance levels across most cancer types, while iClusterPlus showed consistently lower performance when handling multiple subtype distinctions. This pattern was particularly evident in cancers with complex molecular landscapes such as LUAD, which exhibited the most diverse combination patterns and generally lower clustering scores across all methods.

Cancer types demonstrated varying levels of clustering stability. The KIRP showed relatively stable performance across fewer subtype combinations, while THCA and HNSC displayed more variable performance patterns depending on the specific subtype combinations analyzed. The general trend across all cancer types indicated declining performance as the number of subtypes in the combination increased, particularly when including molecularly similar subtypes. These observations show the importance of carefully considering both cancer type-specific molecular characteristics and method selection when designing multi-omics studies involving subtype analysis. The results suggest that optimal study design should prioritize clearly distinct molecular subtypes when possible, while acknowledging that performance may vary significantly based on the specific cancer type and analysis method chosen.

Optimizing omics combinations for clinical features

Evaluating different omics-layer combinations for Subtype, and Pathological Stage (Fig. 5) revealed that integrating GE–ME often led to superior clustering performance, especially for Subtype (Fig. 5a). While GE–MI was also beneficial for Subtype, it did not perform as well for the other test. In contrast, the Pathological Stage (Fig. 5b) benchmarks generally exhibited lower cluster-scores, suggesting that this clinical feature may be more challenging for MOI methods to distinguish. This pattern is reflected in the narrower regions seen in the corresponding panel (Fig. 5b), regardless of the method employed. A closer examination of the combination test showed notable method-specific differences. For instance, SNF, COCA, and NEMO show consistently strong performance when classifying Gender using GE–ME, whereas other approaches (e.g., iCluster, MOFA) may excel in different contexts or for different clinical variables. These findings suggest the importance of matching the method to the data type and clinical interest: the interplay between specific omics layers (e.g., GE–ME vs. GE–MI) and particular clinical variables (e.g., Subtype vs. Gender) can substantially influence clustering success. Consequently, researchers need to evaluate a range of MOI approaches and omics-layer combinations to optimize performance for their specific study goals. The additional results for Gender and Age are available in Supplementary Figure S2.

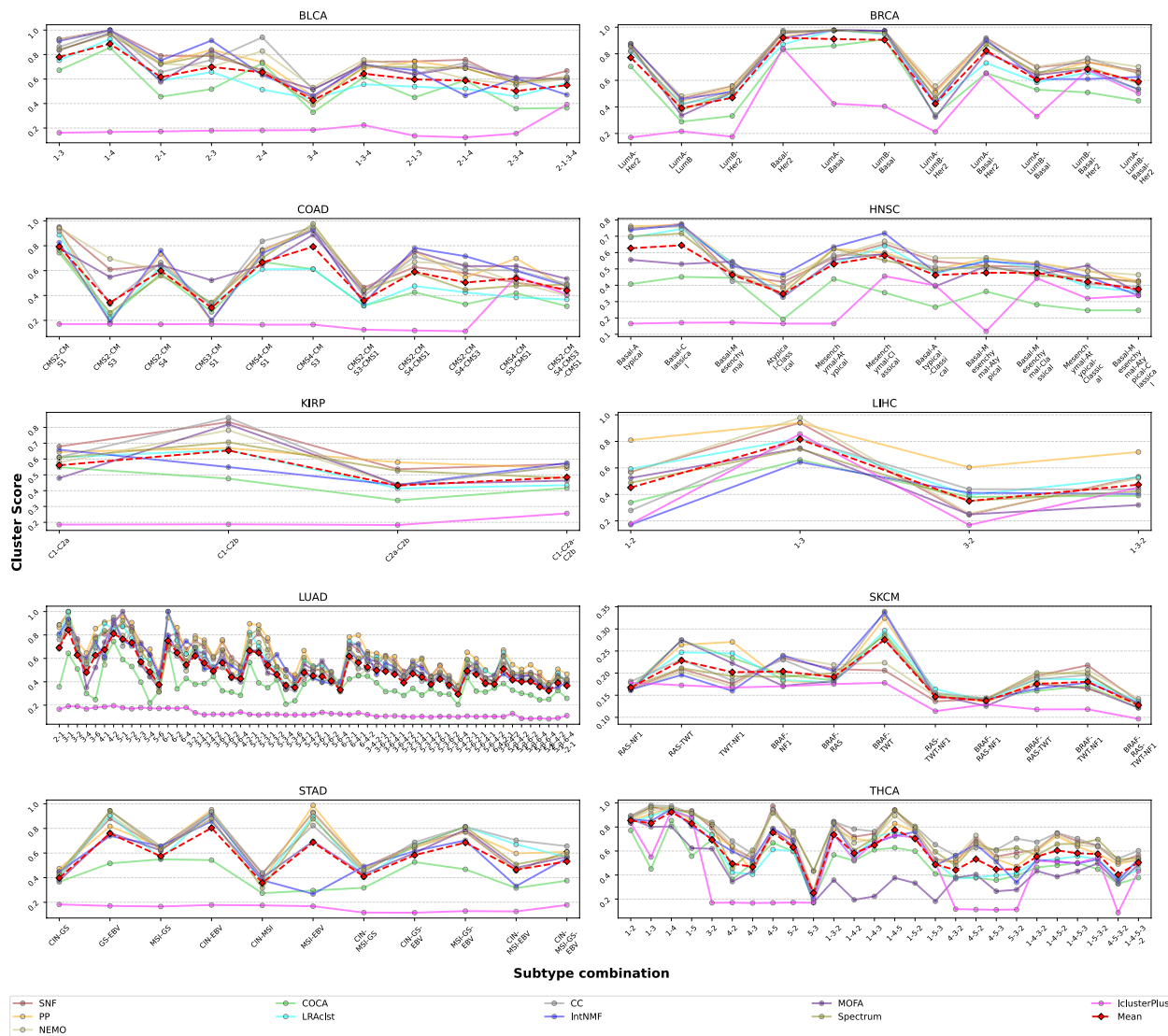


Fig. 4 Method performance across cancer subtype combinations in ten cancer types. Analysis of clustering performance across different cancer subtype combinations in ten TCGA (The Cancer Genome Atlas) cancer types (BLCA, BRCA, COAD, HNSC, KIRP, LHC, LUAD, SKCM, STAD, and THCA). The x-axis represents different subtype combinations, and the y-axis shows cluster scores. The mean performance (red dashed line) indicates overall trends in clustering capability as subtype complexity increases

Comparative performance of ten MOI methods

The comparative analysis of ten MOI methods across different test scenarios revealed distinct patterns of performance strengths (Table 3). SNF demonstrated consistently strong performance in structural tests, leading in sample size (5/10), feature selection (4/10), and subtype combination analyses (5/10). NEMO showed particular strength in clinical feature analysis, achieving top performance in stage-based omics combinations (5/10) and maintaining strong performance across multiple test categories. Method performance varied notably by analytical context. CC showed specific strength in gender-based analyses (4/9), suggesting utility for sex-specific molecular patterns. PP demonstrated robust performance under noise conditions (3/10), while also maintaining

competitive performance in various other categories. Other methods showed more specialized performance profiles. LRAcluster and IntNMF performed well in specific clinical contexts, particularly in gender and age-related analyses, while COCA and Spectrum showed moderate but consistent performance across various test scenarios. This performance heterogeneity emphasizes that method selection should be guided by specific analytical objectives rather than general performance metrics alone. Notably, the balanced performance across top methods in several categories (e.g., three methods sharing top performance in balance analysis) suggests that multiple approaches may be viable depending on specific research contexts. These findings highlight the importance of matching MOI method selection to specific

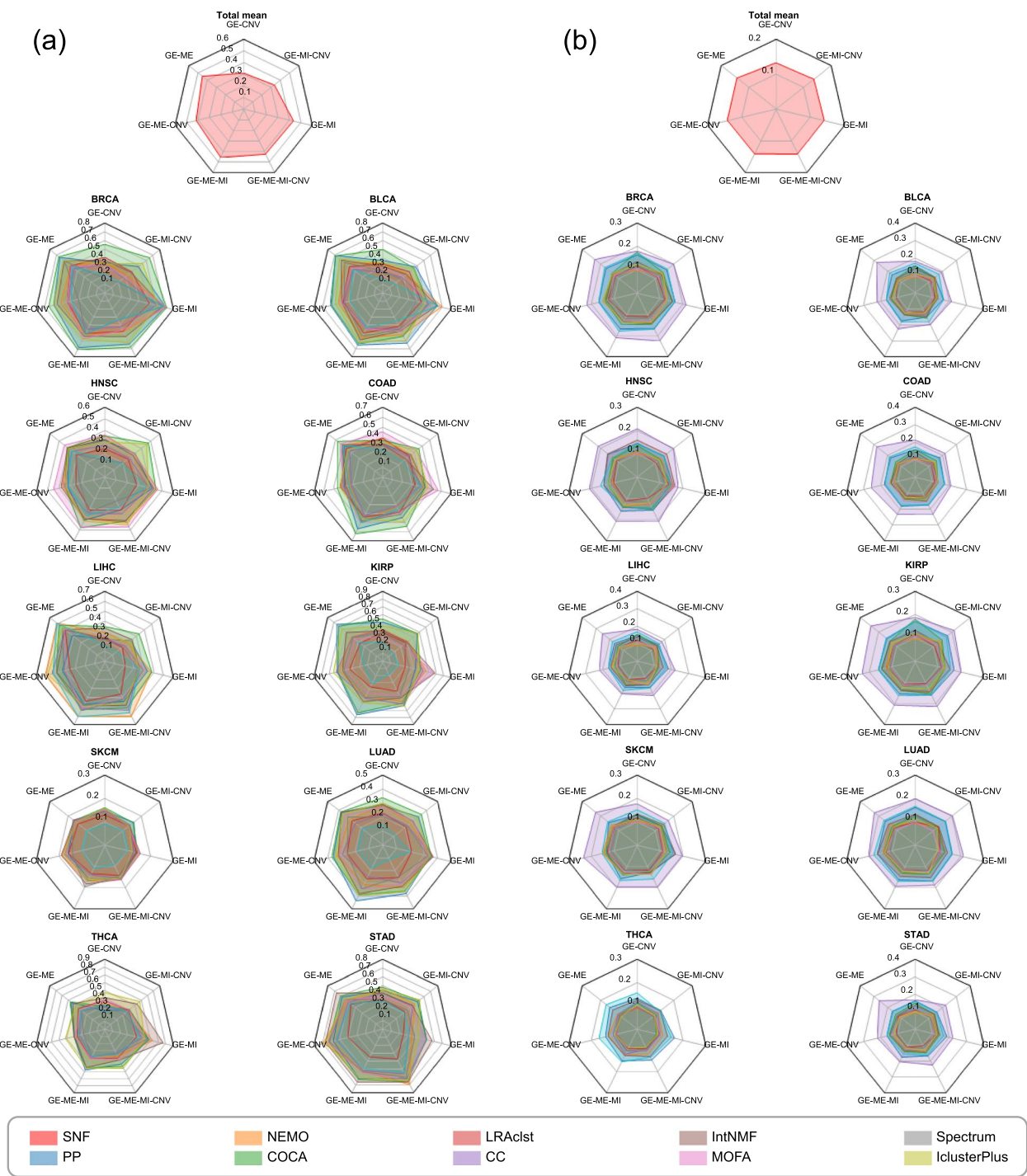


Fig. 5 Performance analysis of multi-omics combinations across clinical features. Radar chart visualization of clustering performance across different omics combinations and clinical features, presented in two sections: summary charts (top) showing mean performance values and detailed method/cancer-specific charts (bottom). **a** Subtype Analysis - Superior clustering performance observed with combinations including GE-ME-MI (Gene Expression-Methylation-miRNA), demonstrating the value of integrating multiple omics layers for subtype identification. **b** Pathological Stage Classification - Generally lower clustering performance across omics combinations, indicating challenges in stage distinction through multi-omics integration

Table 3 Performance ranking of MOI methods across different test scenarios

Test case	Top method	Runner-up	Notable performer
Sample size	SNF (5/10)	NEMO (2/10)	PP (2/10)
Feature selection	SNF (4/10)	NEMO (2/10)	PP, CC (2/10 each)
Balance	SNF, NEMO (3/10 each)	PP (2/10)	CC (1/10)
Noise tolerance	PP (3/10)	SNF, NEMO, CC (2/10 each)	-
Subtype combination	SNF (5/10)	NEMO (2/10)	MOFA, CC (1/10 each)
Omics (Subtype)	NEMO (4/10)	SNF, Spectrum (2/10 each)	CC, IntNMF (1/10 each)
Omics (Gender)	CC (4/9)	LRACluster (2/9)	SNF, COCA, IntNMF (1/9 each)
Omics (Stage)	NEMO (5/10)	PP, CC (2/10 each)	COCA (1/10)
Omics (Age)	NEMO (3/10)	IntNMF, CC, LRACluster (2/10 each)	Spectrum (1/10)

A comparison of ten multi-omics integration methods across various test scenarios. Numbers in parentheses indicate the frequency of achieving the highest cluster score out of total cases (e.g., 5/10 means the method achieved the highest score in 5 out of 10 cancer types). Top method indicates the best performing method in each test scenario, followed by the second-best performer (Runner-up) and other methods showing notable performance (Notable Performer). Test cases include fundamental computational assessments (sample size, feature selection, balance, noise tolerance) and biological analyses (subtype and omics combinations with various clinical features). The gender analysis includes only 9 cases due to the exclusion of one cancer type with extreme gender imbalance

analytical goals and data characteristics in the MOSD framework.

Discussion

MOI has expanded the understanding of cancer biology [3], while practical challenges persist. Factors such as sample size, feature selection, and omics combination selection influence whether analyses yield reliable or misleading insights [59]. Despite advancements in machine learning algorithms, a unifying framework to address variability in data quality, noise, and imbalances has been lacking [60]. The proposed MOSD framework encompasses both computational and biological factors, contributing to standardization of MOI approaches and bridging the gap between theory and practice.

Feature selection plays a significant role in clustering performance—an aspect sometimes overshadowed by focus on algorithms [61]. The findings suggest that feature curation can enhance interpretation and reduce computational overhead, even with modest sample sizes or inherent imbalances. This indicates the importance of considering both quantity and biological relevance of selected features. Benchmark tests revealed that using fewer features (1-10% of the total) generally yielded higher clustering performance, with improvements of up to 34% compared to using all features. This finding

aligns with Pierre-Jean et al. [29]’s observation that feature selection impacts clustering quality, while providing more specific guidance on selection ratios. Sample size analysis demonstrated that 26 samples per class supports robust clustering performance, particularly for reducing variance in results. While absolute performance showed modest improvement with larger sample sizes, the consistency of results improved substantially, indicating that larger cohorts help prevent poor clustering outcomes rather than dramatically improving average performance.

Class balance emerged as another important factor, with results showing that imbalance ratios exceeding 3:1 degraded clustering performance. This finding expands on Mirza [27]’s identification of class imbalance as a challenge in MOI analysis and offers a specific threshold for multi-omics study design. The comparison of raw, normalized, and gene-level preprocessing approaches demonstrated that preprocessing method selection affects clustering outcomes. The gene-level approach recommended by Jung et al. [47] showed advantages in certain contexts, particularly when analyzing molecular subtypes, though effectiveness varied by cancer type and analysis method. For biological factors, certain cancer subtype combinations (e.g., Luminal B-Basal in BRCA) showed high discrimination potential, while others were more challenging to distinguish. This pattern was consistent across different MOI methods, highlighting the relevance of considering molecular distinctiveness of subtypes when designing studies.

Additionally, combining numerous data types does not automatically improve results—particularly when clinical features are difficult to discriminate based on non-linear relationships [62]. The GE-ME pairing frequently emerged as an effective omics combination across multiple cancer types, suggesting that focused integration of complementary omics types may be more effective than comprehensive multi-omics approaches. Method selection also influenced outcomes, with different algorithms showing distinct strengths across various test scenarios. SNF demonstrated consistent performance in structural tests, while NEMO performed well in clinical feature analysis. This method-specific performance variation indicates the importance of matching analytical approaches to specific research questions.

Conclusion

The MOSD framework provides a guide to prioritizing factors such as sample size, feature selection, and omics integration strategy for reliable and interpretable MOI results. The framework incorporates insights from benchmarking of ten MOI methods across ten TCGA cancer datasets, offering evidence-based recommendations for multi-omics analysis. Guidelines emerging from this study include: maintaining approximately 26 samples

per class for reliable results, selecting less than 10% of features to enhance performance, ensuring sample balance under a 3:1 ratio between classes, and keeping noise levels below 30%. For biological factors, distinct molecular subtypes should be prioritized, and omics combinations selected based on the specific clinical features of interest, with GE-ME often providing effective results for subtype identification.

These findings have implications for the design, execution, and interpretation of multi-omics studies, potentially improving efficiency and reproducibility of cancer research. Future research could explore framework adaptation to emerging data modalities and additional factors—such as temporal sampling or single-cell resolution—that may further refine multi-omics analyses. Continued refinement of these guidelines may support more efficient translation of complex molecular data into meaningful clinical and biological insights.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12864-025-11925-y>.

Supplementary Material 1.

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Authors' contributions

I.J. and H.K. conceived the study and designed the experiments. H.K. developed the multi-omics integration methods and performed the data analysis. H.K., E.H., and I.J. contributed to the experimental design and interpretation of the results. H.K., E.H., and I.J. wrote the manuscript. I.J. supported the funding and supervised the study. All authors reviewed and approved the final version of the manuscript.

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Data availability

The data and associated code supporting the findings of this study are publicly available in our GitHub repository at <https://github.com/cobi-git/MOP-ARAM>. The repository includes all necessary materials to reproduce the results presented, along with detailed documentation and instructions for use.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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